



Signature-Based Regime Detection and Enhancements for Lead-Lag Strategies

Design a signature-based regime model and overlay it on a lead-lag baseline to improve Sharpe and reduce drawdowns.

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Data and Pipeline Overview

✱ Regime Detection Input Data (BTC-only)

- ▶ **Instrument: BTC-USD** (1D), 2021-01–2024-06;
- ▶ **Purpose:** build a **market-regime signal** ($-1/0/+1$) via **Signature/MMD** on BTC path segments.
- ▶ **Walk-forward:** rolling clustering $\rightarrow$ prototypes (medoids) on train window $\rightarrow$ map to {bear/neutral/bull} by forward returns; daily regime by **k-vote**.
- ▶ Key parameters: $n_steps \in \{5, 8, 10, 15\}$, $n_paths \in \{8, 12, 16, 20\}$, $window_groups=30$, $n_clusters=3$, $k_vote=5$, $H=30$, $train_end \approx 70\%$ of timeline.

✱ Lead–Lag Matrix Input Data (multi-asset)

- ▶ **Assets:** a **multi-coin basket** (e.g., ETH, XRP, DOGE, NEAR, VET, AAVE, TRX, ATOM, ...)
- ▶ **Rolling window: 30 days (lookback)**
- ▶ **Update frequency: 1D**
- ▶ **Method:** Signature
- ▶ **Entry:** Strength and sign of one asset leads another asset

Regime Detection

✱ Steps:

1) Build trajectory groups (no look-ahead)

- ▶ **Slice the BTC path into sub-paths of length n_steps (offset=0).**
- ▶ **Apply PathTransformer (time normalization + standardisation).**
- ▶ **Stack n_paths consecutive sub-paths into a group → yields a time-ordered bank of groups with $start_time$ and end_time**

2) Measure group-to-group distance (Signature/MMD)

- ▶ **Use AutoEvaluator with RBF-MMD over signature features (dyadic_orders=[2], sigmas=[1]).**
- ▶ **Produce a precomputed distance matrix**

3) Train semantics on an early window ($\approx 70\%$)

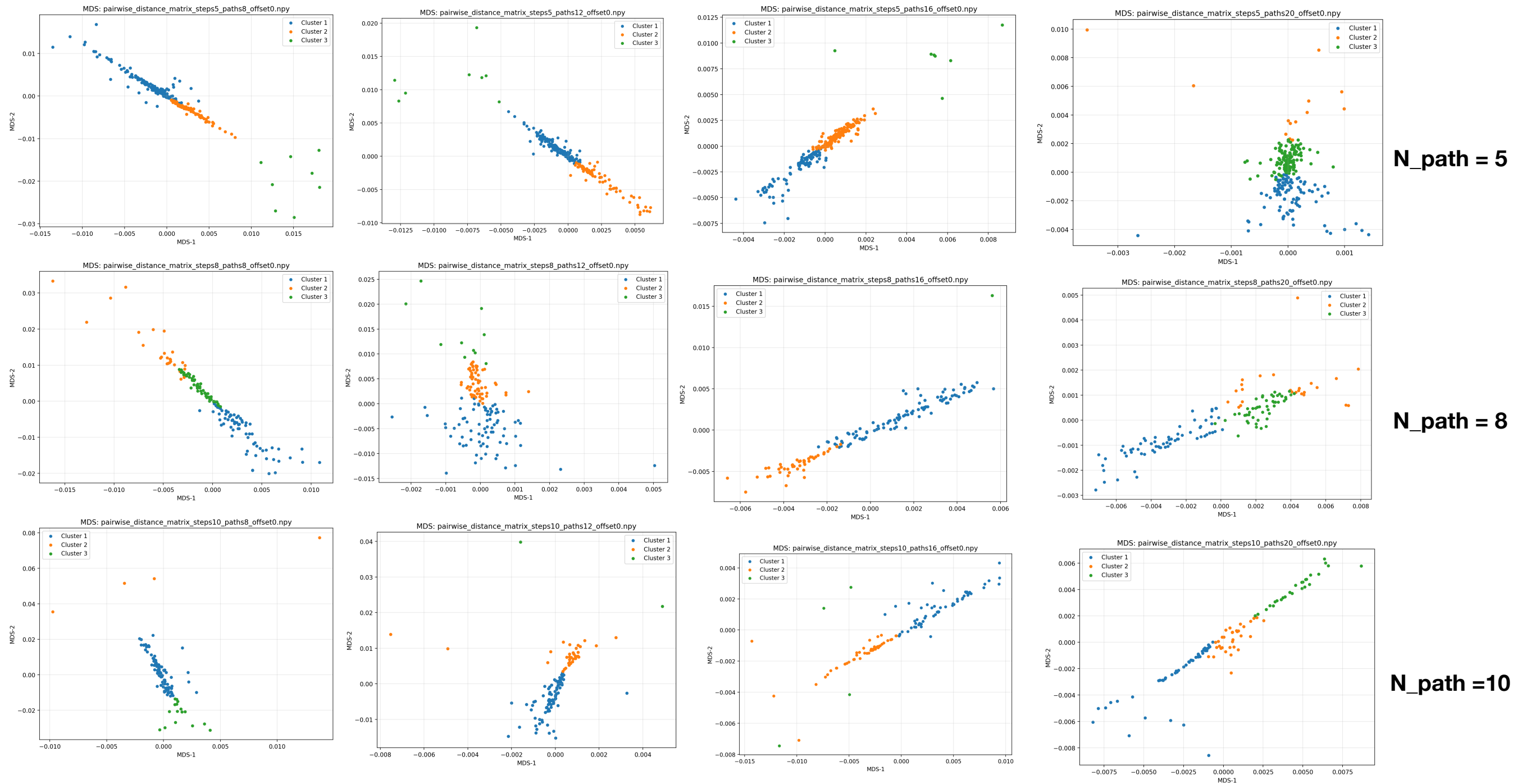
- ▶ **Agglomerative clustering** (complete linkage, metric=precomputed) on the training submatrix
- ▶ **Pick medoids (prototypes).**
- ▶ **Compute forward H-day log returns (start_from="next") per cluster; assign bear / neutral / bull by thresholds.**

4) Walk-forward assignment (no look-ahead)

- ▶ **For each group across the full timeline: nearest medoid → cluster → semantic label.**
- ▶ **Convert to daily regimes via rolling majority vote on the last k completed groups (k_vote).**
- ▶ **Output with labels in $\{-1,0,+1\}$.**

5) Loop for sensitivity analyses: $n_step \in 5,8,10,15; n_path \in 8,12,16,20$

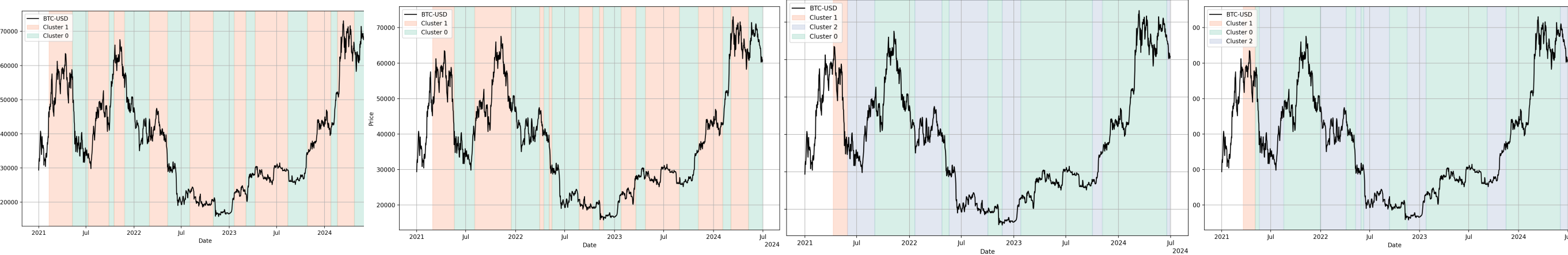
Trajectory Clusters (MDS Projection)



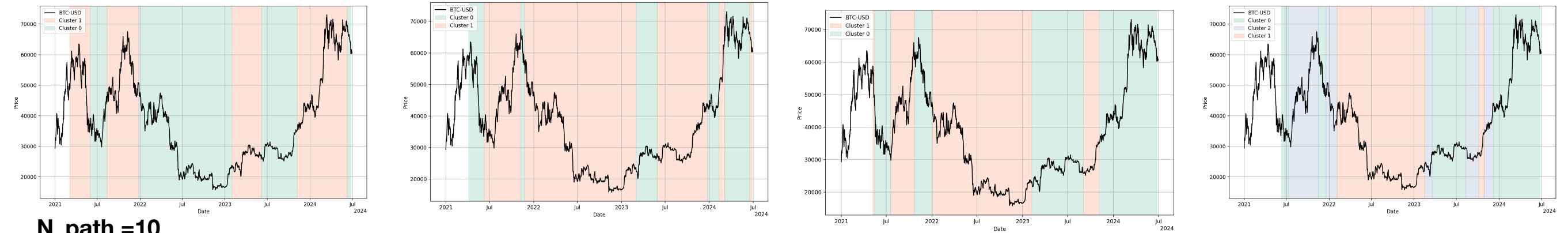
- Each panel shows the 2D MDS projection of the signature-MMD distance matrix for a **specific step/path**
- points lie on an almost **one-dimensional ridge** with **three well-separated clusters**;

BTC Regime (Signature–MMD) — Walk-Forward Spans

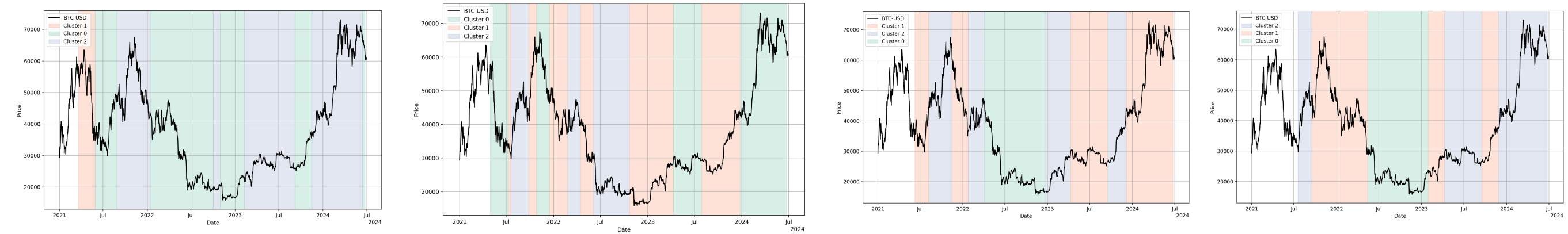
N_path =5



N_path =8

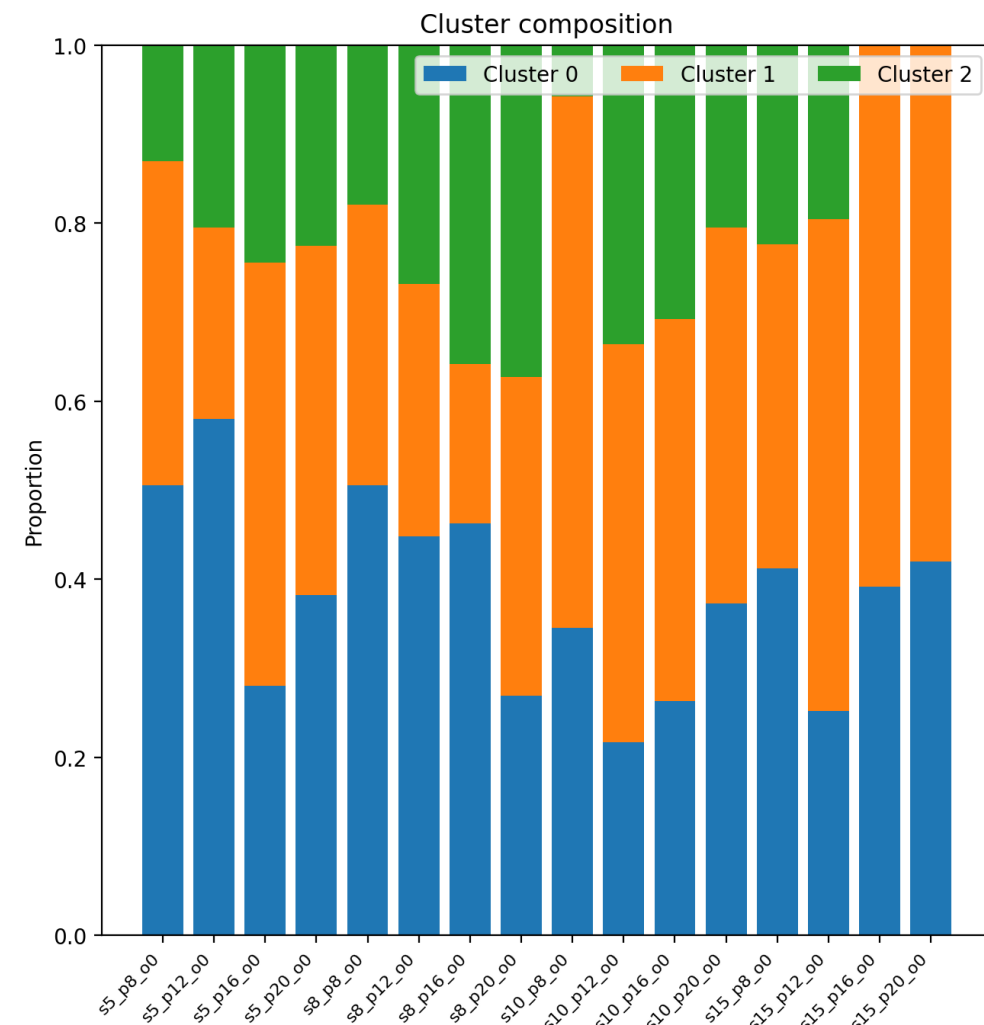
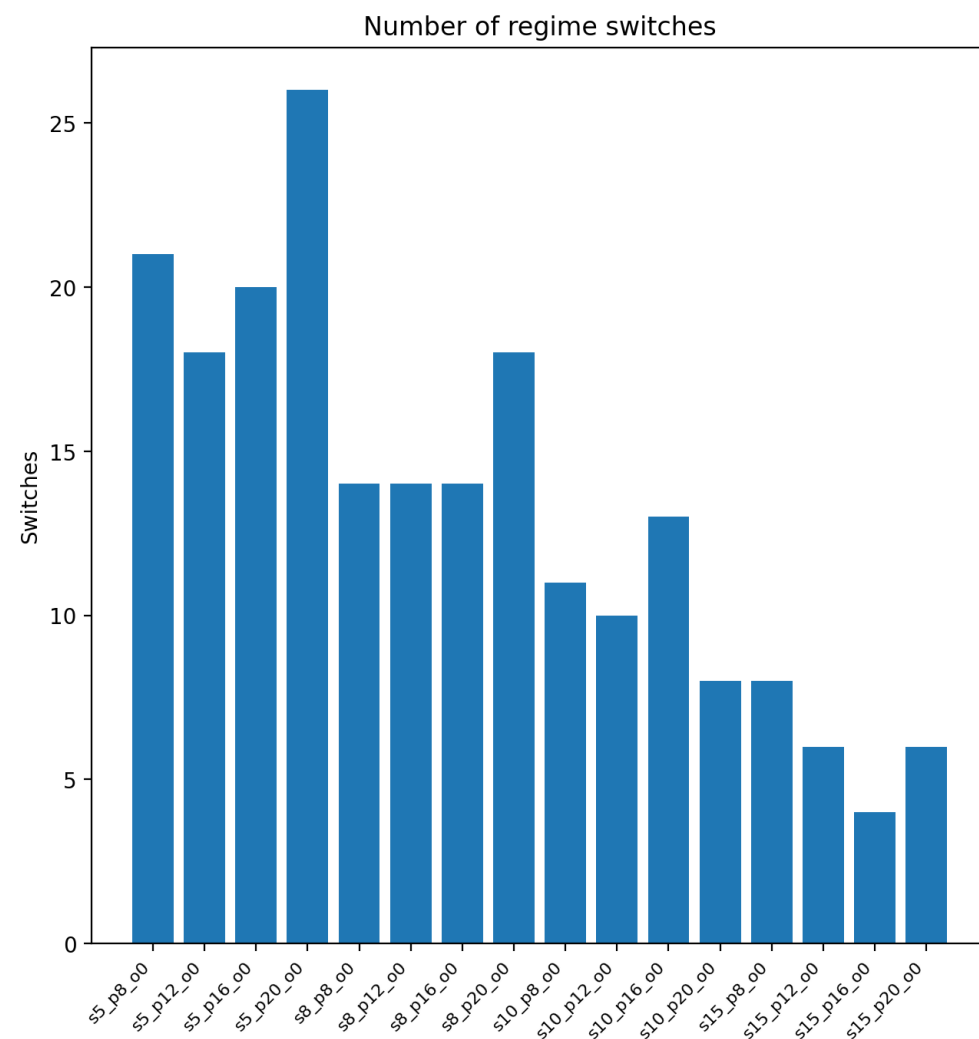


N_path =10



► BTC regimes via signature–MMD distances, trained walk-forward; semantics (bear/neutral/bull) set by forward returns; daily regimes from k-vote.

BTC Regime (Signature-MMD) — Walk-Forward Spans



▶ Short windows (e.g., s5_p*) shows many more regime switches (noisier, more reactive), while larger windows (e.g., s15_p*) yield far fewer switches (smoother, more persistent states).

▶ larger windows (e.g., s15_p*) leads to far fewer Cluster2

Regime-Aware Overlay on Lead-Lag — Method & Baseline

✳ Steps:

1) Baseline (Lead-Lag hedge)

- ▶ Build rolling **lead-lag matrix** from daily log returns (lookback=30, update=1d)
- ▶ **Per day, score assets by row mean; select leaders (positive tail q) & followers (negative tail q).**
- ▶ Trade **next day: long followers / short leaders** (min names filter). $\Rightarrow$ returns

2) BTC Regime (introduced in the previous slides)

3) Overlay logic (our strategy)

- ▶ If neutral and neutral_mode="baseline" $\rightarrow$ pass through baseline.
- ▶ Bear flips the baseline exposure; Bull keeps orientation
- ▶ Trade on $t+1$ with same universe & min_names. $\Rightarrow$ returns

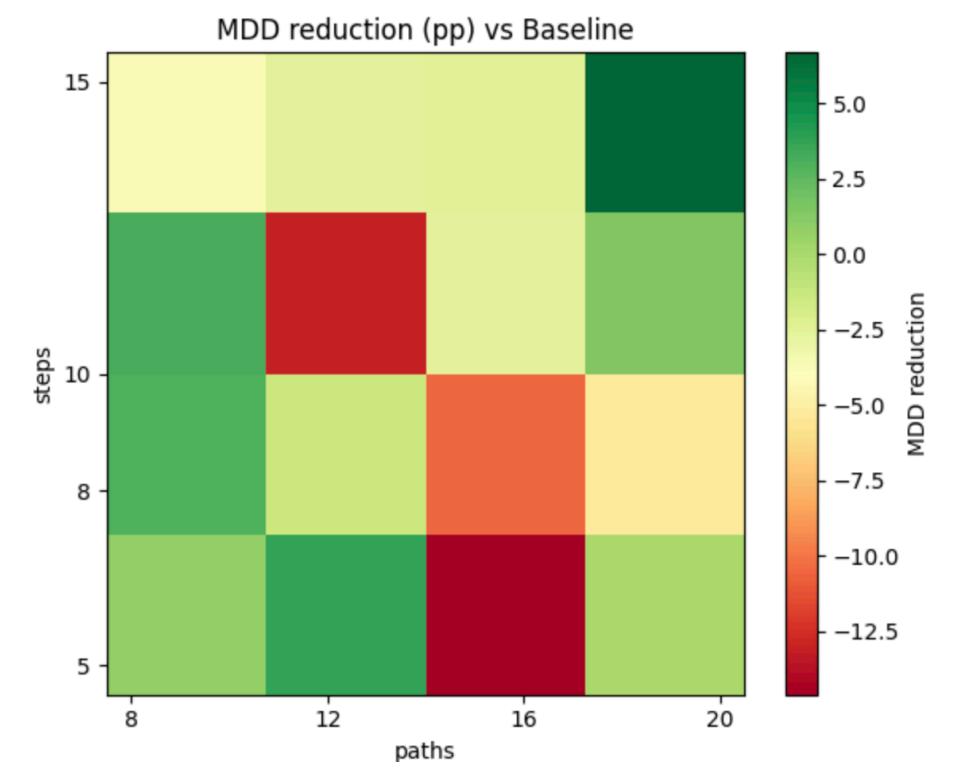
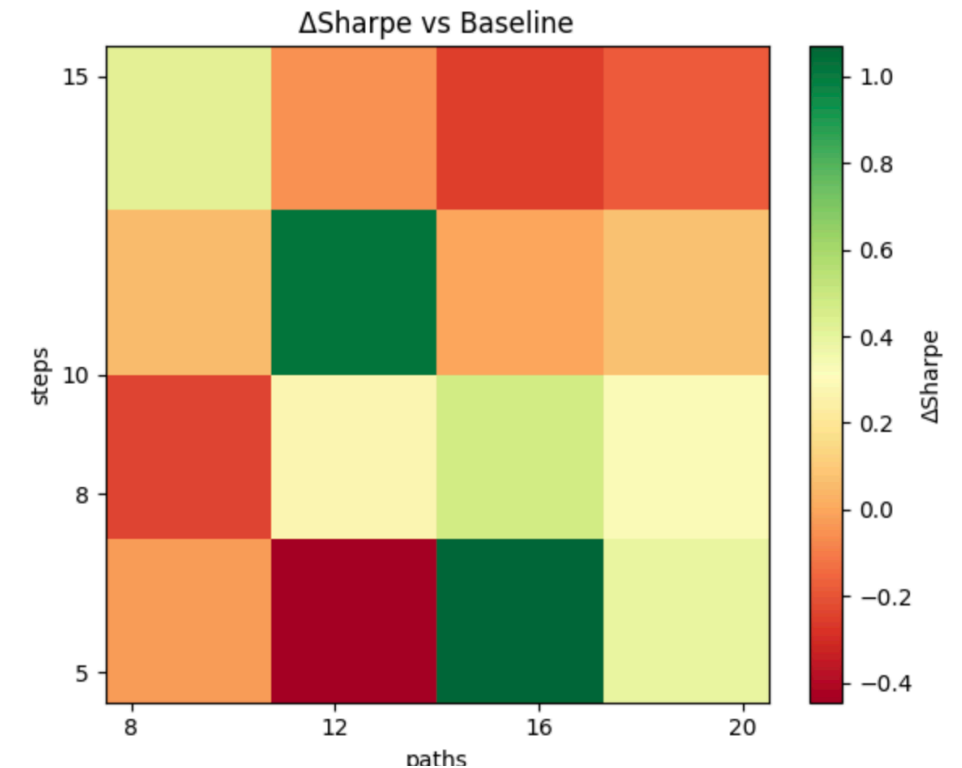
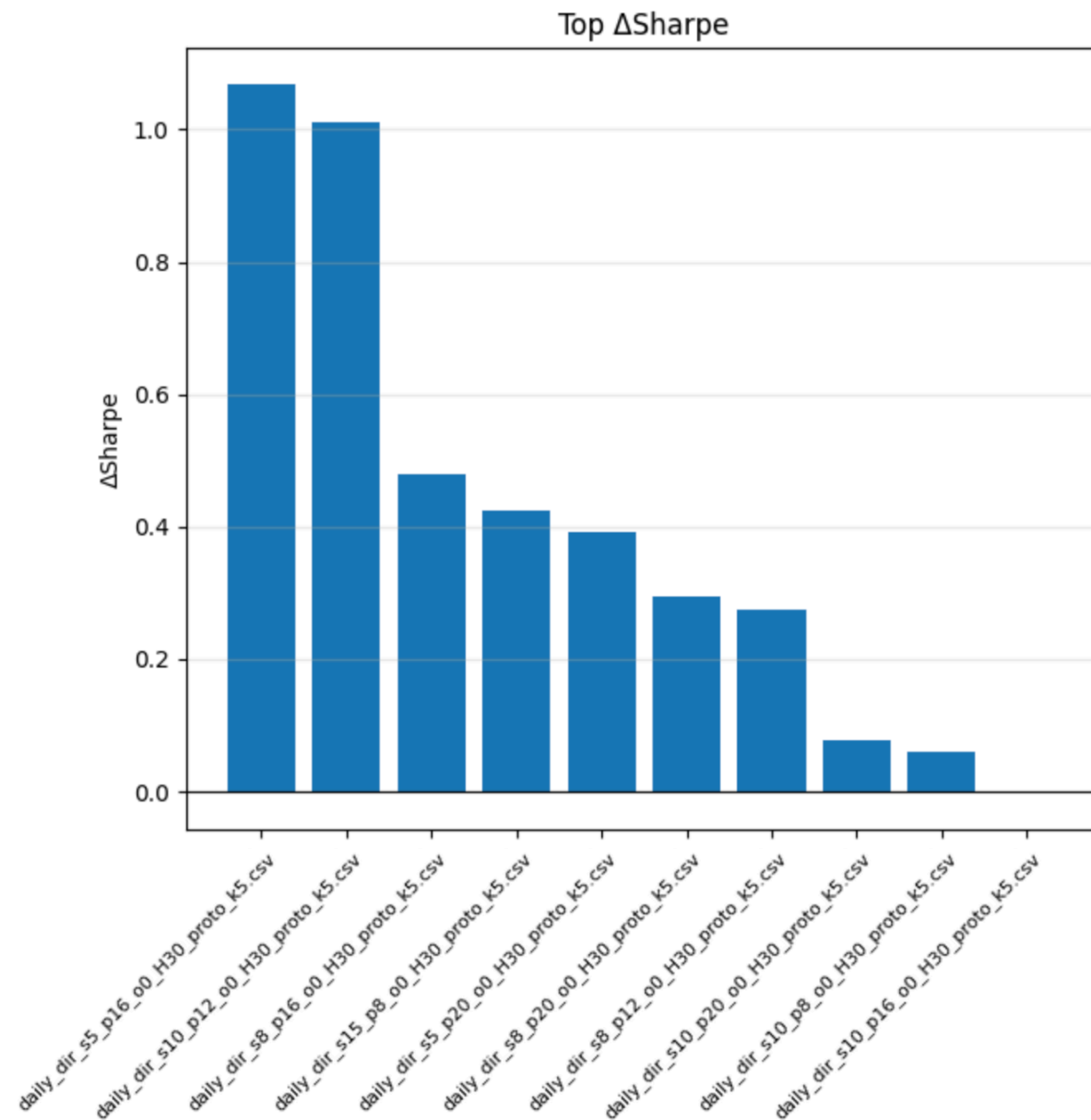
$$r_{\text{overlay}, t+1} = \begin{cases} r_{\text{base}, t+1} & s_t = 0 \text{ and neutral_mode=baseline} \\ \alpha(s_t) r_{\text{base}, t+1} \text{ with sign-aware splits} & s_t \in \{-1, +1\} \end{cases}$$

4) Orientation selection (train only)

- ▶ On the **training segment**, compare Sharpe (or annual return) of ret_overlay vs **its flip**
- ▶ Pick the better orientation; apply that choice to the **full sample**.
- ▶ **Same calendar** (unified daily; missing days = 0), **same $t \rightarrow t+1$ execution**, **same costs**, **same min_names**.

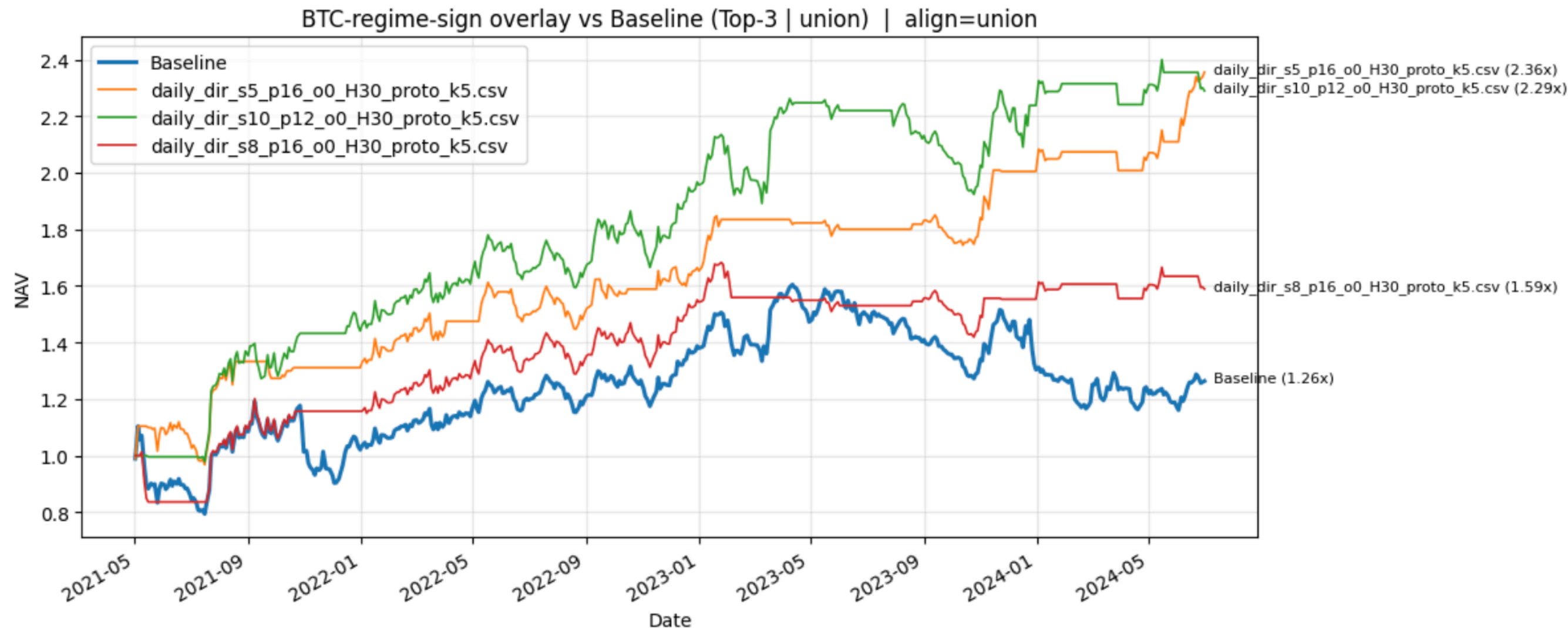
5) evaluation vs Baseline

Delta Sharpe and Max Drawdown(MDD) Reduction



► **Peak Δ Sharpe and the largest max-drawdown reduction (MDD) appears at (steps, paths) = (5,16) and (10,12)**—shorter windows favor return uplift, larger windows tilt toward risk control; overall, many configurations **outperform the baseline**.

Evaluation & Baseline Comparison

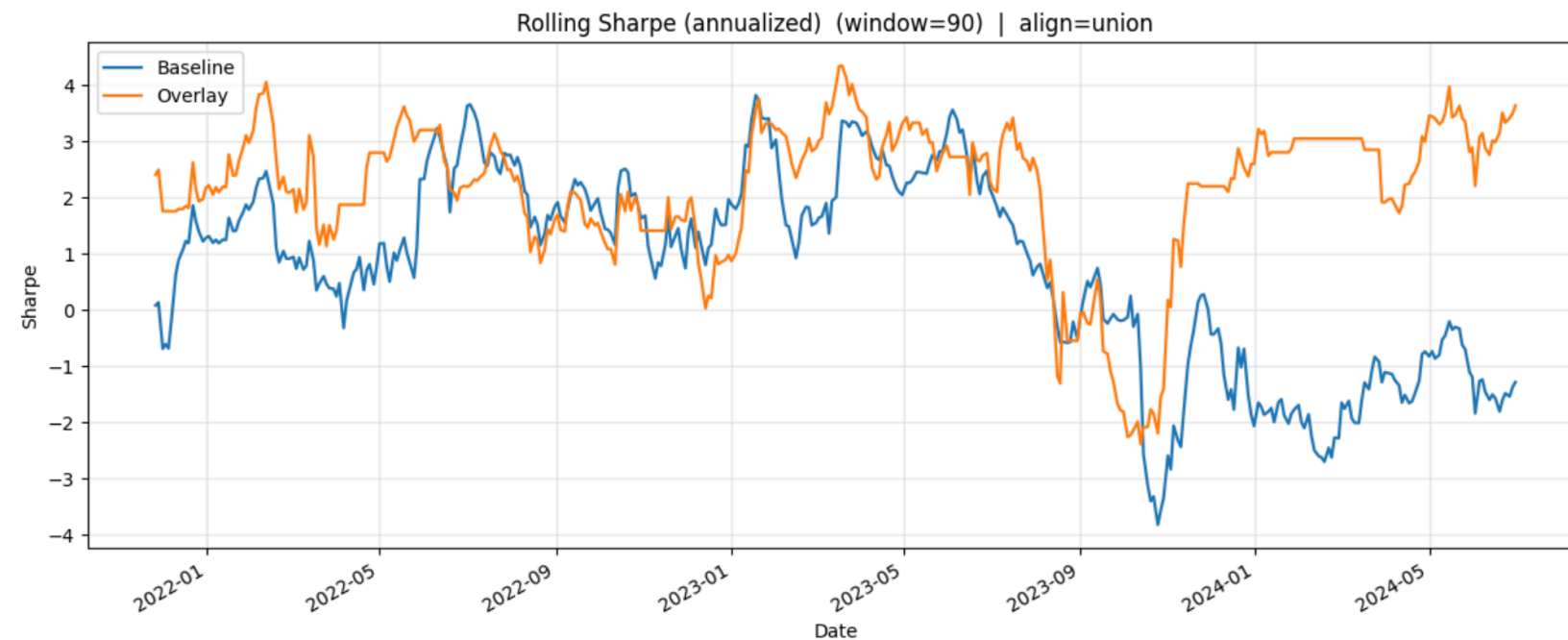


► **Regime overlays materially outperform the baseline** — top configs (s5_p16, s10_p12) finish at ~**2.36x / 2.29x** vs **1.26x** for the baseline.

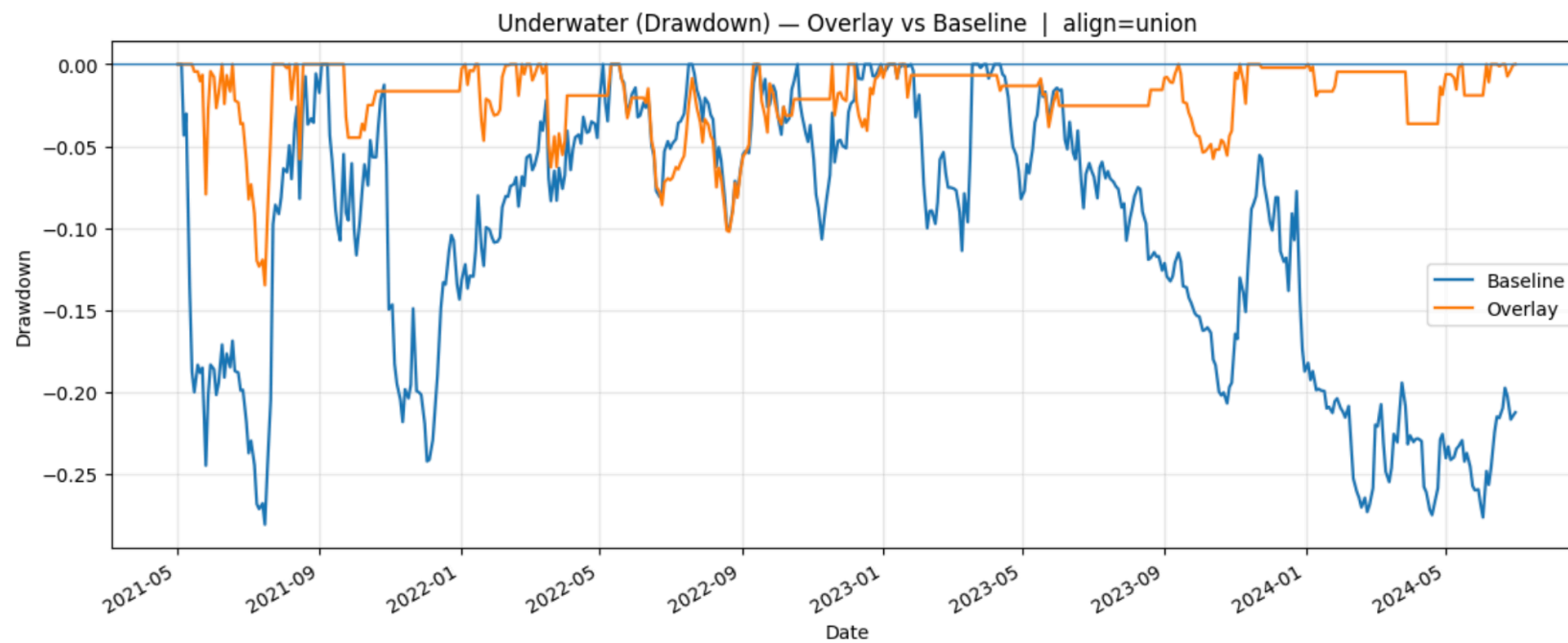
► during the 2023–24 chop, overlay curves are steadier with shallower drawdowns

Rolling Evaluation & Baseline Comparison

$n_Step = 5$ and $n_path = 16$ vs baseline strategy:



Rolling Sharpe: The overlay stays mostly **positive and stable** and **quickly recover** after drop in 2023



Rolling Drawdown: The overlay's drawdowns are **shallower and shorter**

Summary — Regime-Aware Lead-Lag

- ✳ **Goal:** Add a leak-free **regime layer** to a cross-sectional **lead-lag hedge** and test if timing improves risk/return.
- ✳ **Regime:** BTC path → Signature-MMD distances → agglomerative clustering (train window) → **medoids** → semantics by **forward H-day returns** → **daily labels via k-vote** (no look-ahead).
- ✳ **Overlay:** In **bull/bear**, split long/short by the **sign of BTC→asset lead-lag** (and strength filter); in **neutral**, run **baseline**. Train-period chooses flip vs no-flip.
- ✳ **Evaluation:** Unified daily calendar (ann=365), $t \rightarrow t+1$ execution, same universe/min_names; compare **overlay vs baseline with various metrics**.
- ✳ **Headline results:** Top configs s5_p16 / s10_p12 finish $\sim 2.36\times$ / $2.29\times$ vs $1.26\times$ baseline; $\Delta \text{Sharpe} \approx +1$ with largest DD improvement
- ✳ **Next:** Prepare dissertation and discuss other items: adding costs/slippagevolatility targeting, multi-asset regimes...